Homework 10, Week 13 2023.11.29 Due 2022.12.06 补充 AJM† and others

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A particle of mass m performs a one-dimensional movement in the potential : $V(q) = c\, e^{\gamma q}$, with c and γ being both real constant.

Calculate $q(t)$ and $p(t)$ by use of the Hamilton-Jacobi method. Using a generating function of the type $U_2(q, P, t)$

Solution

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A particle of mass m performs a two-dimensional movement in the x-y plane in the potential: $V(x, y) = c\,(x - y)$, with c being a real constant.

Solve the Hamilton-Jacobi differential equation for the characteristic function $W(x, y, P_x, P_y)$. Use the initial conditions

$$x(t = 0) = y(t = 0) = 0, \quad \dot{x}(t = 0) = v_0 > 0, \quad \dot{y}(t = 0) = 0$$

Solution

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Given the linear harmonic oscillator $H = \dfrac{p^2}{2\,m} + \dfrac{1}{2}\, m\, \omega_0^2\, q^2$. Perform in the theory of Hamilton-Jacobi theory a canonical transformation in such a manner the new coordinates (Q, P) are constant of motion. The generating function may be of the type $U(Q, p, t)$.

Calculate $q(t)$ and $p(t)$ with the initial conditions: $p(t = 0) = 0$, $q(t = 0) = q_0 > 0$.

Solution

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Apply the method of the action and angle variables to determine the frequencies of a three-dimensional harmonic oscillator with pairwise different force constants. Does the system perform the periodic motion?

Solution

★ 参照 课件第六章 例题 6.2.09 及文献 **American Journal of Physics 82, 848 (2014)** （已上传）求解 （荣誉课程同学完成）

一个电量为 q 的点电荷在固定于坐标原点，电偶极矩为 $\vec{p} = p_0\, \vec{e}_z$ 的点电偶极子产生的静电场中，

点电荷的静电势能为 $V(r, \theta) = \dfrac{k \cos\theta}{r^2}$，其中 r, θ, ϕ 为球坐标，$k = \dfrac{q\, p_0}{4\,\pi\,\varepsilon_0}$。

(a) 写出体系的拉格朗日量及对应于三个广义坐标 r, θ, ϕ 的拉格朗日方程

(b) 写出对应于循环坐标 ϕ 的运动常数 ℓ，并解释其物理意义

(c) 求点电荷在 $\ell = 0$, $\phi = 0$, $r = r_0$ 条件下的解，并求此解对应的能量 E，其中 r_0 是常数

(d) 写出体系的哈雅方程，并求哈密顿主函数 $S(r, \theta, \phi, E, \alpha_\theta, \alpha_\phi)$

(e) 讨论 E > 0, E = 0, E < 0 三种情况下的解 r(t)，只需给出 r(t) 的函数形式，不必求出常数。

(f) 试判断求解哈雅方程能否得到求解拉格朗日方程 (c) 所求出的解。

若不能，试解释哈雅方程为什么会漏掉这个解。

Solution